

Eventless Payload-Adaptive NMPC for Quadrotors

Anonymous Authors

Affiliation withheld for double-anonymous review

Abstract—A release command does not guarantee physical payload detachment, so switching a controller’s model from an attach/drop signal can be unsafe. We present an eventless payload-adaptive control framework in which a moving horizon estimator (MHE) [1] uses only motor speeds and odometry to publish an atomic, health-gated frame of composite mass, first mass moment $s = m_{PTxy}$, and the inertia increment derived from them to an outer-loop nonlinear model predictive controller (NMPC). The first-moment representation remains well defined as the payload vanishes, while bounded filtering and continuous body-rate scheduling prevent discontinuous model updates. In PX4 [2] SITL with physically attached eccentric payloads, the interface completes release confirmation in 31/31 eligible eventless flights with no observed post-release instability. In 42 % of flights, sufficient no-payload evidence arrives while the vehicle continues the figure-eight, giving a median confirmation delay of 3.46 s. The remaining 58 % lack reliable first-moment evidence in maneuver; after a 12 s timeout, the system enters UNRESOLVED and executes an approximately 3 s brake to hover. The resulting low-dynamic condition restores observability, yielding a median confirmation delay of 16.48 s. A safety ablation shows payload quantity and thrust residual to be individually unreliable in aggressive flight—their fast branch cleared the model prematurely in 3/9 flights; restricting the detector to fresh moment-collapse evidence removed every premature declaration over 12 flights. In five valid paired fault-injection trials, the physical joint was held for 4 s after the release command: the external-event baseline cleared its model before detachment in 5/5 flights, whereas the eventless interface never cleared early in any of its six valid runs and instead retained the loaded model or entered UNRESOLVED until physical evidence arrived. In a separate comparison recorded with an earlier mass-only variant of the estimator, explicit estimation matched an L1-adaptive [3] NMPC in transient tracking, recovered faster (L1 lags by 8 % to 32 %), and retained interpretable physical parameters. These results show that the eventless interface fails safely: when detection is uncertain, it conservatively retains the loaded model rather than clearing it prematurely. They do not establish that the release detector will identify detachment quickly and correctly under all flight conditions.

I. INTRODUCTION

Autonomous aerial delivery and in-flight aerial manipulation subject a multirotor to abrupt changes in its own dynamics. Grasping or releasing a parcel changes three coupled quantities at once: the total *mass* m (which rescales the thrust-to-acceleration map and the hover throttle), the *center of mass* (a thrust that no longer passes through the composite CoM produces a constant trim torque $\tau \approx \Delta r \times f$, significant even for small offsets because $f \approx mg$ is large), and the *rotational inertia* (which slows the attitude response and effectively lowers the inner-loop bandwidth). The first channel is an outer-loop (position/thrust) problem; the latter two are inner-loop (attitude/rate) problems. A controller that models the payload

only as a scalar mass will mistrack or, for sufficiently heavy eccentric loads, destabilize.

Explicit parameter estimation (EKF/MHE of mass, CoM, inertia) is interpretable and can feed model-based constraints, but converges slowly across an abrupt change; *lumped-disturbance adaptation* (L1-adaptive NMPC, INDI) reacts fast but yields no physical model; *learning-based* estimators typically need rich sensing or ground-truth data.

The central question of this work is whether the complete grasp–carry–release lifecycle can adapt its payload model without trusting a commanded or externally supplied payload event. Our answer rests on two observations. First, payload mass and first moment can be estimated continuously, with the inertia increment following algebraically from the mass estimate and a weak geometric prior, without consuming a gripper command, attachment state, or external event notification. Second, **motor speeds are a free physical sensor**: both collective thrust and body torques can be reconstructed from rotor speeds alone, yielding physical residuals that are decoupled from the estimator’s own state. Combining the two yields a *sensing-minimal* estimator that needs no external event signal and no measurement of the attachment geometry.

Contributions.

- 1) An **eventless payload-adaptive NMPC architecture** whose model never consumes an attach/drop notification or treats its own release command as evidence. An atomic, health-gated $m/s/J$ frame drives bounded model updates and continuous body-rate scheduling (§III–IV).
- 2) **Closed-loop lifecycle and safety validation**: 31/31 eligible eventless flights complete eventless release without observed post-release instability, while 23 valid A/B pairs retain the external-event baseline and expose the cost of refusing its shortcut. A five-pair delayed-detachment fault injection directly tests command/plant disagreement: the event baseline clears early in 5/5 paired flights and the eventless interface in 0/6 valid runs. A targeted ablation rejects an unsafe quantity-plus-residual branch that lacks moment evidence and adds an UNRESOLVED fail-safe state (§VI–A).

The runtime estimator consumes only motor speeds and odometry; the sole geometric quantity not identified online is a weak vertical-arm prior r_z , which is structurally unobservable from the thrust-moment channel.

II. RELATED WORK

Agile quadrotor flight. Geometric tracking on SE(3) [4] and time-optimal flight [5] assume known, fixed inertial pa-

rameters, which a payload event invalidates mid-flight.

Aerial manipulation and transport. Aerial robots that grasp, carry and release objects are surveyed in [6]; dynamic aerial grasping, with explicitly modeled gripper dynamics, was demonstrated in [7]. Load transportation, by grasping or cable suspension, is surveyed in [8]. Ours is the transport-and-release subproblem: the payload is rigidly attached, so the controller must handle the resulting change in mass, inertia and centre of mass.

Control under changing payload. Adaptive and robust designs—L1-adaptive NMPC [9], INDI [10]—treat the load as a lumped disturbance and are the standard baselines for payload-carrying flight; L1-NMPC in particular reports large tracking-error reductions under unknown load on aggressive trajectories. A parallel line learns the residual dynamics instead: [11] fits unmodeled aerodynamic effects with Gaussian processes inside the MPC, which improves tracking without yielding a named physical parameter either. Slung-load control addresses the swinging-payload case explicitly [12], exploiting differential flatness of the cable-suspended system; a rigidly attached payload instead changes the vehicle’s own inertial parameters. A related proprioceptive line estimates the external wrench directly with momentum-based observers, enabling collision detection and interaction control [13]. All of these recover a *force/torque*, not a physical parameter: they cannot, e.g., update a thrust-margin constraint or an inner-loop gain schedule from the estimate.

MHE and online parameter estimation. Filter- and optimization-based estimators recover mass, inertia, and geometric parameters [14]–[16]. Standard MHE performance is highly sensitive to the process/measurement weight matrices; neural MHE (NeuroMHE) [17] learns to output these weights online, resolving the bias–variance trade-off but estimating a lumped disturbance force rather than explicit named parameters.

Residual-based change detection. Residual-based fault/change detection (CUSUM and related methods [18]) motivates the evidence checks used by the payload lifecycle state machine. Our positioning: we do not learn dynamics and add no sensing; every deployed signal derives from motor speeds and odometry. This is a deliberate departure from learning-based MPC [19] and the broader safe-learning agenda [20], where a learned component absorbs the model mismatch; here the mismatch is resolved into named parameters that downstream constraints can consume.

III. SYSTEM OVERVIEW AND PROBLEM SETUP

Architecture (Fig. 1). Three layers: Gazebo physics (with a magnetic DetachableJoint gripper producing genuine inertia/CoM changes), the PX4 inner-loop rate controller, and a ROS 2 outer loop (NMPC + MHE). The outer loop publishes a collective thrust and a body-rate setpoint; PX4 tracks the body rate.

Dynamics and parameterization. The NMPC model carries the runtime parameter vector $p = [x_{\text{ref}}, m, \Delta J, c_{xy}]$ (extended with a lumped disturbance only for the L1 baseline).

Thrust acting at the geometric center about the composite CoM produces $\tau_{\text{com}} = [-c_y T, +c_x T, 0]$. The deployed MHE augments both composite mass and the horizontal first mass moment,

$$x_{\text{MHE}} = [x^\top \quad m \quad s_x \quad s_y]^\top \in \mathbb{R}^{16}, \quad (1)$$

$$s = m_P r_{xy},$$

with $\dot{m} = 0$ and $\dot{s} = 0$ within each window ($N = 20$, $\Delta t = 0.1$ s). The published geometry is reconstructed as

$$c_{xy} = \frac{s}{m},$$

$$\mu = \frac{m_B \max\{m - m_B, 0\}}{m}, \quad (2)$$

$$\Delta J_{xx} = \Delta J_{yy} = \mu r_z^2,$$

where r_z is a weak rack prior. The small $O(r_{xy}^2)$ diagonal terms and yaw increment are omitted in the deployed interface. The hover-input baseline is recomputed from the live model and includes the eccentric trim torque, avoiding a stale operating-point bias.

Model-independent physical inputs. We reconstruct the collective thrust $T_{\text{phys}} = k \sum_i \omega_i^2$ and the body torques from rotor speeds and arm geometry. The quadratic coefficient k is a static-thrust constant; it is known to lose validity under large induced inflow during fast translation [21], which we do not model. Feeding the NMPC’s *commanded* thrust back into the estimator would instead close an estimator–controller loop: that command is computed from $\hat{m}$ itself (its cost anchors T near $\hat{m}g$), and any inversion error, saturation, or motor lag between command and rotor is invisible to the estimator, so a mass error can persist self-consistently rather than appear as a residual. The motor-speed reconstruction measures the force downstream of those stages and does not depend on $\hat{m}$.

Thrust-to-throttle inversion. The offboard collective-thrust command is inverted to a normalized rotor command through the motor model $T = k\omega^2$. Provided that the forward thrust reconstruction and inverse command mapping use the same k , steady-state closed-loop tracking is generally not directly affected by its absolute value.

Continuous inner-loop scheduling. Model consistency in the NMPC cannot by itself restore the PX4 rate loop’s lost bandwidth. The deployed path therefore maps the live inertia increment to a body-rate-command scale, low-pass filters it, applies target hysteresis and asymmetric rise/recovery slew limits, and preserves 5% command headroom before the final saturation. It does *not* rewrite PX4 parameters. An attach-time inertia-only bootstrap uses the rack’s 0.30 kg payload envelope for immediate safety, then fades out after persistent coherent MHE evidence; neither payload mass nor s is injected by this bootstrap, and the drop path remains estimate-only.

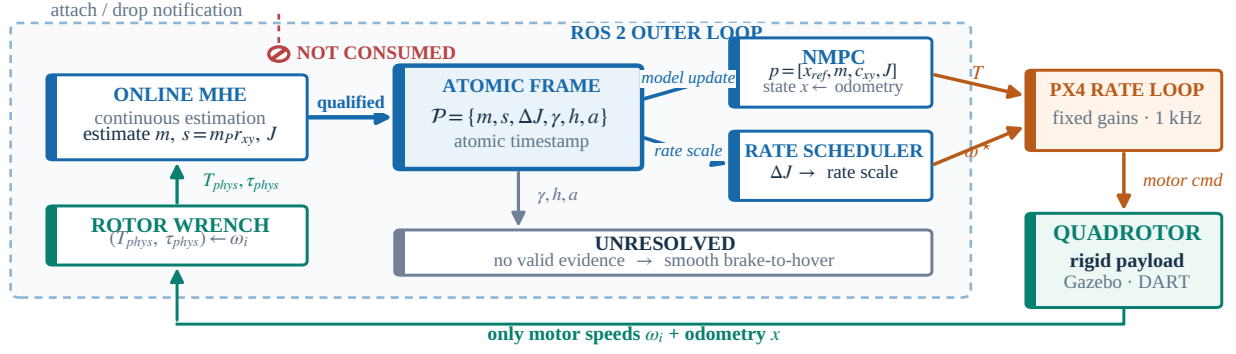


Fig. 1. System architecture and information flow. The full pipeline consumes only motor speeds and odometry; one atomic, health-gated $m/s/J$ frame drives both the NMPC model and continuous rate-command scheduling. No attach/drop notification enters this path.

IV. ONLINE ESTIMATION AND EVENTLESS ADAPTATION

A. Continuous payload interface and autonomous lifecycle

The MHE publishes one atomic frame

$$\mathcal{P} = \{m, s_x, s_y, c_x, c_y, \Delta J_{xx}, \Delta J_{yy}, \Delta J_{zz}, J_{xx}, J_{yy}, J_{zz}, \gamma, h, a\}, \quad (3)$$

where γ is no-payload confidence, h is a health bit and a is solution age. Health requires a successful latest solve and fresh odometry, motor-speed, and known-input streams. The NMPC additionally checks local receive age; an unhealthy or stale frame freezes the last accepted model and breaks the confidence persistence timer. Valid frames pass through a low-pass filter and separate slew limits for m , s , and ΔJ before every NMPC solve.

Payload presence is an estimator-owned state, deliberately distinct from the legacy flag that means “an external attach notification was received.” A hysteretic $\text{EMPTY} \leftrightarrow \text{LOADED}$ machine uses sustained payload-mass evidence, with a directional residual as a fast path, and a separate latch records that the present flight reliably entered LOADED . This latch prevents an empty-flight confidence value from consuming the later drop confirmation and prevents a transient mass dip during a figure-eight from disarming the true release path.

No-payload confidence is the product of smooth empty scores for payload quantity, first moment, and inertia. A nonzero moment therefore vetoes a false empty decision even if the mass estimate touches its lower bound. On accepted release evidence, the estimator switches only the *published target* of s to zero; the optimization state is not overwritten. The output filter consequently produces a continuous decay rather than a geometric step. The controller marks a planned drop complete, resets its L1 state, and finishes recovery only after healthy confidence remains above 0.90 for 1.0s; issuing the gripper command itself does none of these things.

B. Why the first mass moment is the essential state

Estimating CoM offset alone is ill-conditioned at release: physically $s = mpr_{xy} \rightarrow 0$, whereas r_{xy} itself is undefined as

$m_P \rightarrow 0$. In an earlier mass-only formulation, the optimizer could suppress a phantom eccentricity only by driving total mass below the bare-airframe value; 14 of 55 post-release samples then pinned at the lower bound. Augmenting s supplies the rotational channel directly and permits $c_{xy} = s/m$ without knowing the grasp-dependent horizontal offset. A frozen-window inertia coefficient prevents the optimizer from using instantaneous mass as a free inertia knob, while the weak r_z prior retains the dominant parallel-axis term. This structure is the minimum one that simultaneously represents payload existence, trim moment, and roll/pitch inertia without an external attachment geometry measurement.

The release detector studies three physically distinct sources: (i) collapse of $\rho = \|s\|/\|s\|_{\text{ref}}$ below 0.30, where the loaded reference is formed from a healthy, low-dynamic, in-envelope interval and frozen before aggressive flight, with an absolute $\|s\|$ bound used only as a sanity check; (ii) payload quantity, in which mass and algebraically derived inertia count as *one* source; and (iii) a directional thrust-residual edge with finite validity. Within this detector no single source releases. The slow path requires quantity-empty and moment-collapse evidence for five MHE frames. The release detector uses a signed two-window thrust-residual difference, $\Delta T = \bar{T}_{\text{post}} - \bar{T}_{\text{pre}}$, with 0.3s half-windows. We evaluated two fast residual branches: a *quantity-residual* branch paired a normal 0.9N residual with quantity-empty evidence, while a *moment-residual* branch paired a longer-persistent residual with moment collapse. A recorded figure-eight transient reached -1.69N , larger in magnitude than the -1.47N step of the minimum supported payload, so amplitude alone cannot separate the cases. Online and offline evaluation call the same pure decision function. The randomized validation of Sec. VI-A rejects the quantity-residual branch; the detector therefore fires only on moment-collapse evidence (slow or moment-residual) and treats an unresolved timeout as unknown, never as proof of an empty vehicle. Two further routes remain in the final interface and are reported with their frequencies in Sec. VI-A: a mass-domain path counted only in low-dynamic flight, and controller confirmation on

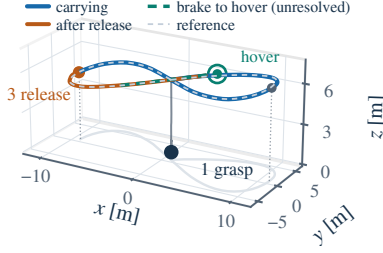


Fig. 2. Measured mission trajectory (simulator ground truth), two 0.15 kg/2 m/s flights of the same batch: grasp (1), lift, then carry along the stereo figure-eight (blue) against the nominal reference (pale dashed). After the release command (3) one flight clears the moment vote in maneuver and keeps tracking (orange); the other declares UNRESOLVED at its 12 s timeout and brakes to a hover (teal), confirming at 16.40 s. The two paths agree to within 0.12 m (median) before the timeout; loaded tracking RMS is 0.053 m, and 0.060 m after release.

the published no-payload confidence. These routes lie outside the moment-gated detector and do not reinstate the rejected branch: the mass-domain path does not accumulate in the aggressive maneuvers where that branch fired prematurely, and the confidence route multiplies in an absolute first-moment score. Neither uses the frozen-reference ratio, so confirmations through them are not counted as validation of that detector. Moment evidence must also originate from a fresh, successful MHE solve: the re-anchoring routine deliberately initializes s to zero after five consecutive solver failures, so that initialization is invalid evidence rather than a physical collapse.

V. SIMULATION SETUP

PX4 SITL with Gazebo [22] (Harmonic), an x500 quadrotor, and a magnetic DetachableJoint gripper that produces genuine inertia and CoM changes.

The physical DetachableJoint gripper attaches a separate rigid body and therefore couples mass, inertia and CoM together. It supports the lifecycle, baseline, and safety claims in the paper; no wrench-emulated mass-step result is retained.

Task sequence (Fig. 2): approach (position control) → NMPC hand-off and warmup hover → controlled descent → active grasp enable → lift → figure-eight → phase-triggered release. We state the NMPC scope explicitly: it governs the grasp hover and the post-event adaptation phase; the approach controller is not part of the claim. Headless batch infrastructure with a GCS-heartbeat stand-in; the physical-effectiveness timestamp $t = 0$ is defined by T_{phys} crossing the baseline, and in-band is defined as ± 0.08 kg or 15 % of the step, whichever is larger.

Real-time budget. Both optimizers are acados [23] SQP solvers with merit backtracking and the HPIPM [24] QP solver. On one desktop core shared with Gazebo/PX4 SITL, the deployed 13-state NMPC ($N = 20$, 20 Hz) solves in 1.3 ms at the median and 2.2 ms at the 99th percentile ($n = 1009$), i.e. 2.6 % and 4.4 % of its 50 ms period. The deployed 16-state first-moment MHE ($N = 20$, 10 Hz) solves in 9.2 ms at the median, 24.6 ms at the 90th and 59.2 ms at the 99th percentile, with a 72.8 ms maximum ($n = 3740$, first 20 frames per flight

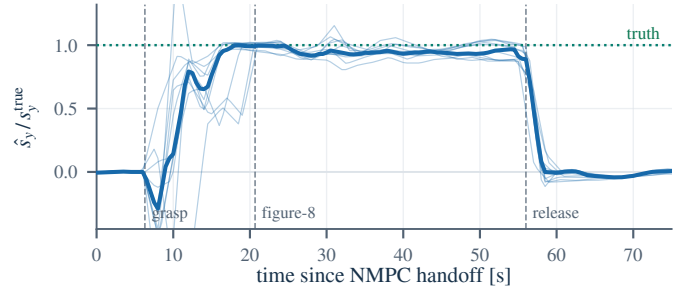


Fig. 3. First mass moment against ground truth (0.15 kg at 2 m/s, $n = 11$, each normalised by its own truth). Truth is reconstructed offline as $m_P r_y$ from the simulator-measured grasp offset and never reaches the estimator, which runs with no horizontal attachment geometry. $\hat{s}_y$ is zero before grasp, rises with no attach notification, holds near truth through the figure-eight, and decays to zero after release. Across all 31 eligible flights the carry-phase estimate is biased toward zero (slope 0.85; median magnitude underestimate 10.8%, same direction in 31/31; largest at 0.30 kg, median 15.7%); the release criterion uses the ratio $\|s\|/\|s\|_{\text{ref}}$ and is insensitive to this scale bias.

discarded), against a 100 ms period—roughly $7\times$ the 1.4 ms median of the historical mass-only estimator. This meets the deadline in simulation, but the tail margin is narrow, so we make *no* onboard-headroom claim; that requires measurement on the target computer. Setpoints are published at 50 Hz by interpolating the latest prediction.

VI. RESULTS

Provenance. The lifecycle cohort uses the final 16-state first-moment interface. The baseline matrix was recorded with the earlier mass-only 14-state MHE and supports only the bounded baseline comparison stated below. In Table III, *truth* gives the NMPC the attachment geometry measured in simulation, *online* replaces the CoM offset with a motor-torque-inversion estimate, and *ll* is the L1-adaptive baseline.

A. Eventless grasp-release lifecycle

The controller issues the gripper-release command itself, but deliberately does not use that command to clear its payload model: a commanded release is not proof of physical detachment. The proposed interface consumes only healthy, fresh $m/s/J$ frames and completes a release only after its no-payload confidence remains above 0.90 for 1.0 s. We report every eventless flight in the final campaign that met the pre-registered eligibility criteria: successful MHE payload evidence and survival to the release command. Exclusions are reported below and are not absorbed into the completion result.

a) Paired external-event baseline: We retain the pre-registered paired ablation rather than selecting arm B alone after the comparison. Arm A is an external-event baseline: it treats the release command as the detachment event and immediately resets payload geometry and inner-loop scaling. Arm B is the proposed eventless interface above. A pair enters this comparison only when both arms satisfy the same eligibility criteria. There are 23 such pairs (8/7/8 across the three conditions); both arms complete 23/23 releases with no observed post-release instability (Table I). Arm A has zero

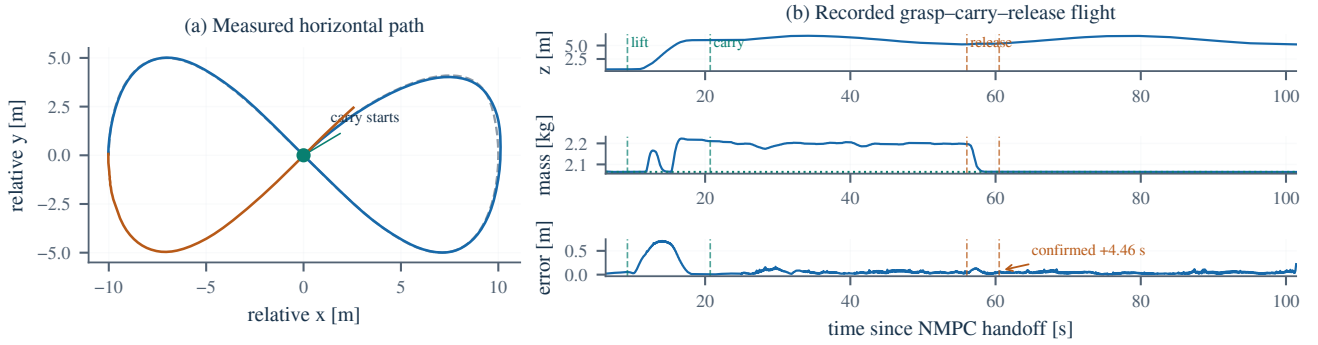


Fig. 4. One eventless flight from the final batch (0.15 kg, 2 m/s). Left: the measured horizontal path (blue while carrying, orange after release) against the nominal reference including its amplitude ramp (grey dashed). Right: recorded height, controller-consumed mass estimate, and position-error norm through grasp, lift, carrying, release command, and confirmation. This flight confirms in maneuver, 4.46 s after the command; it illustrates one execution and does not replace the cohort statistics in Table II.

TABLE I

PAIRED EXTERNAL-EVENT ABLATION. ARM A CLEARS THE MODEL FROM THE COMMAND BY CONSTRUCTION; ARM B REQUIRES ESTIMATOR EVIDENCE. THE LAST ROW INJECTS A COMMAND/PHYSICAL-DETACHMENT MISMATCH; “EARLY” MEANS MODEL CLEARING BEFORE THE DETACHABLEJOINT IS REMOVED. THE ONE VALID BUT UNPAIRED ARM B RUN OF THAT ROW ALSO DID NOT CLEAR EARLY (0/6).

condition	pairs	A compl.	B compl.	early clear	A/B
0.15 kg, 4 m/s	8	8/8	8/8	–	
0.30 kg, 4 m/s	7	7/7	7/7	–	
0.15 kg, 2 m/s	8	8/8	8/8	–	
nominal pooled	23	23/23	23/23	–	
0.15 kg, 2 m/s + 4 s hold	5	5/5	5/5	5/5 / 0/5	

confirmation delay by definition, whereas arm B waits for physical evidence. This comparison therefore quantifies the cost of refusing the event shortcut; it is not evidence that arm A detects physical detachment.

To exercise the mismatch that motivates the paper, a fault injector received the normal release command but retained the physical DetachableJoint for a median 4.01 s. Twelve arm-runs were attempted; one arm A run never armed PX4, leaving five valid pairs and one valid but unpaired arm B run. In all five pairs arm A cleared its payload model at the command—a median 4.01 s before physical detachment. Arm B never cleared before detachment in any of its six valid runs, the unpaired one included: it entered UNRESOLVED in 5/6 and then confirmed a median 12.45 s after detachment, while the remaining run confirmed 3.00 s after detachment without entering UNRESOLVED. Within the five pairs, median position-error peaks during the four-second hold were 0.071 m (A) and 0.068 m (B), so we claim a semantic safety distinction, not a tracking advantage. All eleven valid runs had an independent Gazebo DETACHED state record.

The eventless interface completed autonomous release confirmation in 31/31 eligible flights with no observed post-release instability (Table II). The two-sided 95 % Clopper-Pearson interval is [88.8 %, 100 %], so this result demonstrates closed-loop feasibility rather than a zero failure rate.

The confirmation delay is *bimodal*, and the two modes are

two distinct control paths rather than a spread within one path. In 42 % of flights the no-payload confidence crosses threshold while the vehicle is still flying the figure-eight and confirmation completes in 3.46 s (median; one such flight is shown in Fig. 4). In the remaining 58 % no moment vote is available in maneuver— $\hat{s}$ has drifted under sustained excitation or the frozen reference never formed—so the system reaches its 12 s timeout, declares the payload state UNRESOLVED, and executes a 3 s smooth brake to hover; deceleration alone restores observability and confirmation then completes at 16.48 s (median). Reporting one median across both paths would be misleading, so Table II separates them.

In 26 of the 31 flights the payload-presence machine made exactly two transitions—one load declaration at grasp and one empty declaration at release—with no spurious carrying-phase transition. In the five remaining flights, all of which confirmed in maneuver, the estimator’s presence machine never declared EMPTY within the logged window (over 40 s past confirmation); confirmation rested on the published no-payload confidence alone, as detailed below. Four of the 31 load declarations were raised by the estimator’s own directional thrust-residual evidence rather than by sustained mass evidence; no external grasp signal was consumed. The published first moment decayed continuously rather than stepping to zero (Fig. 3), confirming that recovery used the intended estimator interface rather than the controller’s release command.

b) Safety boundary: The 31/31 batch validates the interface once its release evidence is accepted; it does not validate a universally robust detector. We therefore tested three physically distinct evidence sources: payload quantity, directional thrust residual, and collapse of the normalized first moment. A fast branch combining quantity and residual without moment evidence declared release prematurely in 3/9 randomized flights during a 4 m/s figure-eight. Loaded-flight mass estimates could remain below the nominal empty threshold for up to 48 s, while a dynamic thrust excursion reached -1.69 N, exceeding the -1.47 N weight change of the smallest payload. Neither source is therefore reliable on its own in this regime.

TABLE II

EVENTLESS RELEASE COMPLETION AND CONFIRMATION DELAY (ALL ELIGIBLE EVENTLESS FLIGHTS). DELAY RUNS FROM THE RELEASE COMMAND TO THE FRAME WHERE THE NO-PAYLOAD CONFIDENCE HAS HELD FOR THE FULL 1.0s, SO IT INCLUDES THAT HOLD. THE TWO DELAY COLUMNS ARE DISTINCT CONTROL PATHS AND MUST NOT BE POOLED.

condition	n	compl.	delay median [IQR]	
			in-maneuver	via brake-to-hover
0.15 kg, 4 m/s	9	9/9	2.78 s [2.76, 2.90]	16.56 s [16.46, 16.60]
0.30 kg, 4 m/s	11	11/11	7.08 s [6.96, 7.50]	16.40 s [15.76, 16.60]
0.15 kg, 2 m/s	11	11/11	3.46 s [3.16, 4.36]	16.48 s [16.40, 16.50]
pooled	31	31/31	3.46 s [2.86, 4.46]	16.48 s [16.36, 16.60]

Pooled completion 100 %, two-sided 95 % Clopper–Pearson CI [88.8 %, 100 %]; overall delay max 16.70 s. The share of flights confirming only after the brake-to-hover maneuver is 56 %, 64 % and 55 % (rows in order) and 58 % pooled; the complement confirms in maneuver.

We rejected that branch: the estimator’s unified release detector now fires only on fresh, successful moment-collapse evidence (the slow and moment–residual paths); the legacy mass-domain route that remains is quantified below. Ambiguous cases enter UNRESOLVED; a timeout does not clear the model or reset adaptive state. In a subsequent 12-flight randomized validation, no flight declared release before the command: all 11 confirmed releases passed through moment-gated paths (9 moment–residual, 2 slow) with a median detector delay of 1.9 s, and one remained safely unresolved. That validation preceded the frozen moment reference.

c) Release paths in the final batch: The moment-gated detector is not the only route to confirmation in the final code, and the 31-flight batch of Table II exercised three. The controller confirms release when the published no-payload confidence—the product of mass, first-moment and inertia “empty” scores, where the moment score uses an absolute bound on $\|s\|$ rather than the frozen-reference ratio—holds above 0.90 for 1.0 s on healthy, fresh frames. (i) In 19/31 flights (9/9, 1/11 and 9/11 in the rows of Table II) the moment-gated detector declared release first and the confidence followed. (ii) In 7/31 flights, all at 0.30 kg, the flights that reached UNRESOLVED, the frozen moment reference was never established because its low-dynamic, in-envelope sampling window never filled, so the ratio test never ran; the confidence crossed threshold during or shortly after the brake to hover, and the estimator’s mass-domain path ($m_P < 0.030$ kg for 20 frames, counted only in low-dynamic flight) declared release 0.9–3.1 s later. (iii) In 5/31 flights (three at 0.30 kg, two at 2 m/s), all confirmed in maneuver, the reference was likewise never established and the estimator never declared release; confirmation rested on the confidence alone. No flight in any path declared release before the command. The supported safety conclusion is therefore bounded: the randomized validation supports the moment-gated detector, while the final batch shows that 12/31 confirmations did not pass through it. Broader detector robustness remains outside the present claim.

The cohort is conditional on successful MHE payload evidence and survival to the release command. Of 40 eventless flights, 31 were eligible: six crashed before release and

TABLE III

THREE-WAY METHOD $\times$ SCENARIO MATRIX ($n = 5$): POS_ERR PEAK [M] / RECOVERY [S].

cell (mass/ecc)	truth	online	l1
0.2 / 0.05	0.806 / 4.28	0.768 / 3.70	0.821 / 4.26
0.2 / 0.10	0.794 / 3.62	0.810 / 3.92	0.812 / 4.24
0.3 / 0.05	0.834 / 3.56	0.837 / 3.72	0.883 / 4.90
0.3 / 0.10	1.011 / 3.64	0.876 / 3.88	0.891 / 4.54

three lacked MHE payload evidence. The available logs cannot distinguish physical capture failure from estimator non-recognition in those three cases. Confirmation delay is a median 3.46 s when it clears in maneuver and 16.48 s when the vehicle must first brake to hover.

Across the paired campaign there were 81 attempts: arm A contributed 41 attempts (30 eligible, 11 pre-release crashes), and arm B contributed 40 (31 eligible, six pre-release crashes and three payload-evidence failures). Only the 23 pairs in which both arms were eligible enter Table I; all 31 eligible B flights enter Table II. These are different estimands and are not substituted for one another. Tail-window tracking is not compared: arm B brakes to hover after an unresolved timeout, while arm A continues the figure-eight.

B. Baselines and ablations

L1-NMPC three-way matrix (Table III). {truth, online, l1} $\times$ {0.2, 0.3} kg $\times$ {0.05, 0.10} m, $n = 5$, 60/60 runs stable, zero divergence. (i) Within-cell transient peaks agree to 0.05 m; L1 is higher in the two low-eccentricity cells ($p = .008, .012$ uncorrected, not surviving correction over eight tests), so we claim *no* transient advantage. (ii) Against the deployed *online* estimate, **L1 recovers 8 % to 32 % slower** (median 16 %; significant in three of four cells); the mechanism is that its compensation low-pass cutoff ω_c is bounded by closed-loop stability (0.5 stable, 1.0 unstable on this platform), a bandwidth wall that explicit estimation does not face. (iii) L1 yields no interpretable m/c_{xy} and still requires the same class prior for gain scaling; it is a translational 3-DoF augmentation driven by T_{phys} (the commanded thrust self-excites).

VII. LIMITATIONS AND FUTURE WORK

All system-level evidence is presently SITL-only. Hardware migration replaces simulated motor speeds with ESC telemetry and must re-identify thrust/torque coefficients, noise floors, timing, and gripper compliance before free flight. The vertical arm r_z is unobservable from the thrust-moment channel and remains a weak rack prior; the s_x /pitch channel is noisier than s_y /roll. The deployed diagonal inertia approximation also drops the small $O(r_{xy}^2)$ and yaw-inertia terms.

Release-detector boundary. First-moment augmentation is now part of the default path. The safety ablation and three final-batch confirmation paths are reported in Sec. VI-A; 31/31 completions do not establish distribution-wide detector robustness. In 12/31 flights the healthy, low-dynamic, in-envelope reference window never filled after the LOADED declaration, so

the moment-ratio test never ran. Out-of-envelope $\hat{s}$ samples are excluded from moment votes but still reach the controller's c_{xy} through filtering and slew limits. Reliable reference formation therefore remains necessary for a robust autonomous release detector.

All reported release batches explicitly enable the release-evidence path. Its built-in default is off, under which every flight ends UNRESOLVED, so deployment requires an explicit enable plus startup validation of the configuration.

First-moment drift. Sustained figure-eight excitation can increase $\|\hat{s}\|$ from 0.012 kg m to 0.031 kg m; withholding moment evidence then delays confirmation. Deceleration restored the vote within 3.3 s to 3.4 s in the affected test flights. Together with missing-reference cases, this yields the brake-to-hover delays in Table II. An online counter of first-moment envelope exceedances correlates with this outcome at 4 m/s (in-maneuver confirmation 7/8 for zero exceedances versus 1/12 otherwise; Fisher $p < 0.001$), but misses all six delayed 2 m/s cases. Earlier exit from the maneuver remains unimplemented and requires validation.

Selection and pre-release failures. The system results condition on successful payload evidence and survival to the release command. Of 40 eventless flights, 31 met these criteria; six crashed before release and three lacked MHE payload evidence. Two crashes occurred before the figure-eight at 0.15 kg/2 m/s, whereas four were loaded figure-eight divergences at 0.30 kg/4 m/s, an operating boundary of that condition. Their grasp offsets ($|r_y| = 0.07$ m to 0.14 m) and pre-maneuver roll trim reconstructed from rotor speeds (0.20 N m to 0.40 N m against a 0.5 N m limit) lie within the range of the eleven surviving flights at that condition, so attachment-induced torque saturation does not explain them; their mechanism remains unidentified. We therefore claim no improvement in grasp, lift, or figure-eight stability; attachment observability and pre-release instability require separate experiments.

A sustained vertical up/downdraft remains intrinsically unfounded with a payload change on a collective-thrust-only residual; wind is not evaluated in the reported cohort.

Simulator fidelity. (i) gz-sim Harmonic cannot change a link's inertia at runtime in a way the physics adopts (issue #2733), so every payload change is produced by the DetachableJoint platform, which attaches a separate rigid body (§V). (ii) The payload is rigidly attached; gripper compliance and pendulum modes are not modeled. Neither affects which signals the estimator consumes.

VIII. CONCLUSION

We presented an eventless payload-adaptive NMPC interface driven by motor speeds and odometry. An atomic, health-gated $m/s/J$ frame supports continuous model and body-rate adaptation without treating release commands as detachment evidence. The interface completed 31/31 eligible eventless SITL flights without observed post-release instability, conditional on attachment and pre-release survival. The safety ablation rejects quantity-plus-residual clearing without

moment evidence; the final batch also exercised mass-domain and confidence-only confirmation paths. Under a 4 s delayed-detachment fault, the external-event baseline cleared its model before physical release in 5/5 valid pairs, whereas the eventless interface did so in 0/6 valid runs and remained conservative until estimator evidence arrived. These results demonstrate interface feasibility, not a universally robust detector. Reliable moment-reference formation and hardware validation remain open.

REFERENCES

- [1] C. V. Rao, J. B. Rawlings, and D. Q. Mayne, "Constrained state estimation for nonlinear discrete-time systems: stability and moving horizon approximations," *IEEE Transactions on Automatic Control*, vol. 48, no. 2, pp. 246–258, 2003.
- [2] L. Meier, D. Honegger, and M. Pollefeys, "PX4: A node-based multithreaded open source robotics framework for deeply embedded platforms," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2015, pp. 6235–6240.
- [3] N. Hovakimyan and C. Cao, *$\mathcal{L}_1$ Adaptive Control Theory: Guaranteed Robustness with Fast Adaptation*. SIAM, 2010.
- [4] T. Lee, M. Leok, and N. H. McClamroch, "Geometric tracking control of a quadrotor UAV on SE(3)," in *IEEE Conference on Decision and Control (CDC)*, 2010, pp. 5420–5425.
- [5] P. Foehn, A. Romero, and D. Scaramuzza, "Time-optimal planning for quadrotor waypoint flight," *Science Robotics*, vol. 6, no. 56, p. eabh1221, 2021.
- [6] A. Ollero, M. Tognon, A. Suarez, D. Lee, and A. Franchi, "Past, present, and future of aerial robotic manipulators," *IEEE Transactions on Robotics*, vol. 38, no. 1, pp. 626–645, 2022.
- [7] J. Thomas, G. Loianno, J. Polin, K. Sreenath, and V. Kumar, "Toward autonomous avian-inspired grasping for micro aerial vehicles," *Bioinspiration & Biomimetics*, vol. 9, no. 2, p. 025010, 2014.
- [8] D. K. D. Villa, A. S. Brandão, and M. Sarcinelli-Filho, "A survey on load transportation using multirotor UAVs," *Journal of Intelligent & Robotic Systems*, vol. 98, no. 2, pp. 267–296, 2020.
- [9] D. Hanover, P. Foehn, S. Sun, E. Kaufmann, and D. Scaramuzza, "Performance, precision, and payloads: Adaptive nonlinear MPC for quadrotors," *IEEE Robotics and Automation Letters*, vol. 7, no. 2, pp. 690–697, 2022.
- [10] E. J. J. Smeur, Q. Chu, and G. C. H. E. de Croon, "Adaptive incremental nonlinear dynamic inversion for attitude control of micro air vehicles," *Journal of Guidance, Control, and Dynamics*, vol. 39, no. 3, pp. 450–461, 2016.
- [11] G. Torrente, E. Kaufmann, P. Foehn, and D. Scaramuzza, "Data-driven MPC for quadrotors," *IEEE Robotics and Automation Letters*, vol. 6, no. 2, pp. 3769–3776, 2021.
- [12] K. Sreenath, N. Michael, and V. Kumar, "Trajectory generation and control of a quadrotor with a cable-suspended load — a differentially-flat hybrid system," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2013, pp. 4888–4895.
- [13] T. Tomić, C. Ott, and S. Haddadin, "External wrench estimation, collision detection, and reflex reaction for flying robots," *IEEE Transactions on Robotics*, vol. 33, no. 6, pp. 1467–1482, 2017.
- [14] J. Svacha, J. Paulos, G. Loianno, and V. Kumar, "IMU-based inertia estimation for a quadrotor using Newton–Euler dynamics," *IEEE Robotics and Automation Letters*, vol. 5, no. 3, pp. 3861–3867, 2020.
- [15] V. Wüest, V. Kumar, and G. Loianno, "Online estimation of geometric and inertia parameters for multirotor aerial vehicles," in *IEEE International Conference on Robotics and Automation (ICRA)*, 2019, pp. 1884–1890.
- [16] C. Böhm, M. Scheiber, and S. Weiss, "Filter-based online system-parameter estimation for multicopter UAVs," in *Robotics: Science and Systems (RSS)*, 2021, paper 087, DOI 10.15607/RSS.2021.XVII.087.
- [17] B. Wang, Z. Ma, S. Lai, and L. Zhao, "Neural moving horizon estimation for robust flight control," *IEEE Transactions on Robotics*, vol. 40, pp. 639–659, 2024, open source: RCL-NUS/NeuroMHE.
- [18] M. Basseville and I. V. Nikiforov, *Detection of Abrupt Changes: Theory and Application*. Prentice-Hall, 1993.

- [19] L. Hewing, K. P. Wabersich, M. Menner, and M. N. Zeilinger, "Learning-based model predictive control: Toward safe learning in control," *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 3, pp. 269–296, 2020.
- [20] L. Brunke, M. Greeff, A. W. Hall, Z. Yuan, S. Zhou, J. Panerati, and A. P. Schoellig, "Safe learning in robotics: From learning-based control to safe reinforcement learning," *Annual Review of Control, Robotics, and Autonomous Systems*, vol. 5, pp. 411–444, 2022.
- [21] M. Bangura and R. Mahony, "Nonlinear dynamic modeling for high performance control of a quadrotor," in *Australasian Conference on Robotics and Automation (ACRA)*, 2012.
- [22] N. Koenig and A. Howard, "Design and use paradigms for Gazebo, an open-source multi-robot simulator," in *IEEE/RSJ International Conference on Intelligent Robots and Systems (IROS)*, vol. 3, 2004, pp. 2149–2154.
- [23] R. Verschuere, G. Frison, D. Kouzoupis, J. Frey, N. van Duijkeren, A. Zanelli, B. Novoselnik, T. Albin, R. Quirynen, and M. Diehl, "acados—a modular open-source framework for fast embedded optimal control," *Mathematical Programming Computation*, vol. 14, no. 1, pp. 147–183, 2022.
- [24] G. Frison and M. Diehl, "HPIPM: a high-performance quadratic programming framework for model predictive control," in *IFAC-PapersOnLine (21st IFAC World Congress)*, vol. 53, no. 2, 2020, pp. 6563–6569.